

Optimization-Based Decision Support for Equitable Expansion of Federally Qualified Health Centers in Chicago

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March 2026

Abstract

This report presents a spatial optimization framework for equitable expansion of Federally Qualified Health Centers in Chicago. We document four critical gaps in the current system: the absence of location evaluation tools for decision-makers, an obsolete Index of Medical Underservice (IMU) formula, a limited Medically Underserved Area (MUA) designation process, and unmapped referral networks despite federal mandates. We describe how we plan to address each one. We report on the data infrastructure and the interactive dashboard we have been developing, and outline what our optimization model can deliver: design of catchment areas and larger health zones, facility location analysis, and redesigned IMU formulas. The report includes a roadmap for the next steps and the idea of a follow-up research.

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1 The Problem and Motivation

Chicago’s primary care safety net is operated largely through Federally Qualified Health Centers (FQHCs), community-based clinics that receive federal funding through HRSA’s Health Center Program to serve low-income and uninsured populations regardless of ability to pay. In Chicago, 25 parent organizations operate 166 primary care FQHC sites (obtained from 213 total sites by removing shelters, school-based outreach, and administrative offices), managed under 1,363 grants awarded by HRSA’s Bureau of Primary Health Care between 2016 and 2026 [13]. Nationally, in 2024 more than 32.4 million people used HRSA-funded health centers through 139.4 million total visits; about 90% of these patients had incomes at or below 200% of the federal poverty level, including 25.1 million uninsured, Medicaid, and Medicare patients [13]. Federal appropriations have grown from \$554 million in 1990 to over \$6 billion today [7]. Since 2014, the number of medically underserved people has nearly doubled from 56 million to over 100 million; a study commissioned by NACHC found that over 100 million people in the United States may not have access to a usual source of primary care due to a shortage of providers in their community [19].

HRSA regulates FQHCs through the Health Center Program Compliance Manual [4], which imposes requirements that are spatial planning problems:

- R1. Catchment area design** (Ch. 3): Health centers must define and annually review the boundaries of the catchment area to be served such that the area’s size ensures services are available and accessible to the residents of the area promptly; boundaries must eliminate, to the extent possible, barriers resulting from the area’s physical characteristics, its residential patterns, its economic and social groupings, and available transportation.
- R2. Needs assessment** (Ch. 3): Centers must assess unmet need using available health resources in relation to the size of the area and its population, including appropriate ratios of primary care physicians, along with health indices (infant mortality), economic factors (poverty rate), and demographic factors (population 65+), the exact four components of the IMU score.
- R3. Accessible locations** (Ch. 6): Sites must be accessible to the patient population relative to where this population lives or works, considering access barriers and distance and time taken for patients to travel to or between service sites.
- R4. Hospital referral relationships** (Ch. 8): Each center must develop an ongoing referral relationship with one or more hospitals to assure continuity of care within the catchment area.
- R5. Collaborative relationships** (Ch. 14): Centers must establish and maintain collaborative relationships with other health care providers within the catchment area, local hospitals, and specialty providers to reduce the non-urgent use of hospital emergency departments.

Each of these requirements involves spatial decisions such as where to draw boundaries, how to measure accessibility, which hospitals to partner with, yet HRSA provides no quantitative tools for making them. Health centers are left to determine these on their own, and the results are uneven. Below we identify four critical gaps in the current system.

Gap 1: Decision makers have no way to evaluate a location. When an organization applies for an HRSA New Access Point grant, it must justify *why this location*. But there is no tool to answer that question rigorously. How many underserved residents are within 30 minutes by transit? What is the current capacity of nearby facilities? Will the new site relieve overloaded neighbors or duplicate existing coverage? Nationally, Chang et al. [6] showed that new FQHCs certified between 2007 and 2014 were much less likely to be in high-poverty areas than existing ones (20.5% vs

31.4%). Lee et al. [16] found that FQHC patient volume is 3.27 times higher in areas with at least one FQHC than in areas with none, confirming that spatial availability directly drives utilization. Despite a large body of work on healthcare facility location models, no published study has applied spatial optimization specifically to FQHC site selection. The current system responds to where grant applications originate, not where need is greatest.

Gap 2: The IMU score is outdated. HRSA designates Medically Underserved Areas (MUAs) using the Index of Medical Underservice (IMU), a composite of four indicators: provider-to-population ratio, infant mortality, poverty rate, and percentage aged 65+. Each is converted to a 0 to 25 scale and summed. The total IMU score ranges from 0 to 100, where lower values indicate greater underservice. Areas scoring $\text{IMU} \leq 62$ qualify as medically underserved. But this formula dates from the 1970s. Ricketts et al. [22] argued nearly 20 years ago that the method no longer captures modern healthcare access barriers. It omits insurance coverage, a primary barrier to care. It omits transportation access, often the main factor in whether a clinic is reachable. It omits chronic disease burden (diabetes, hypertension, obesity), the conditions FQHCs actually treat. And it uses a hard binary cutoff: two neighboring tracts with IMU scores 63 and 62 are considered not eligible and eligible, respectively. Moreover, a tract scoring 61 and a tract scoring 30 are treated identically, even though their needs are vastly different. Ricketts et al. [22] also proposed a data-driven replacement, and HRSA issued a Federal Register proposed rule in 2008 to adopt it, but the rule was never finalized. No alternative scoring system has been developed at the research level since.

Gap 3: The MUA designation process has significant limitations. To obtain an MUA designation, an organization must submit an application to HRSA proposing a rational service area, typically a group of census tracts. The applicant draws the boundary, not HRSA [4]. HRSA then computes the IMU score for the proposed area. This means boundaries can be *manipulated* to push IMU below 62, there is no systematic process to determine which groupings of tracts *should* be designated, and communities without an organization capable of navigating the federal application never get designated at all. Ricketts et al. [22] argue that this applicant-driven process systematically excludes communities that lack an organization capable of navigating the federal designation system. Milano [18] found a similar pattern statewide in New Jersey: 1,139 census tracts scored medium or high risk on a composite access index, but only 248 held MUA designation. In Chicago, Kim et al. [15] found that 87 census tracts meet IMU eligibility but lack formal designation, predominantly on the South Side. Between 2009 and 2012, 12 clinics closed while 68 opened, yet MUA-eligible but undesignated tracts received only 3 new FQHCs compared with 41 in designated tracts. The resulting FQHC rate in undesignated tracts was only 15.5 per 100,000 poor, versus 40.8 in designated areas. The designation gap is built into the process.

Gap 4: HRSA requires referral agreements but provides no network analysis. Ch. 8 requires that each health center develop an ongoing referral relationship with one or more hospitals. Ch. 14 requires collaborative relationships with local hospitals and specialty providers to reduce the non-urgent use of hospital emergency departments. But which hospital should a given clinic partner with? Are there communities where patients have no viable referral path to specialist care? Research shows only 34.8% of referrals in the U.S. result in completed specialist visits [20], and the failure rate is worse in underserved areas. No city-level physician referral network has been constructed for Chicago or any other US city, existing work [3] operates only at the national level.

These gaps are connected. In 2005, the Chicago Department of Public Health [23] documented that the ratio of safety net capacity to medically needy population varied tenfold across regions, from 0.4

in the Northwest to 5.1 in the Central region. There were 484,120 low-income uninsured Chicagoans, and only 35% received community-based care; 312,395 had no medical home. The South region saw visits decline even as the number of sites increased, suggesting that *where* facilities are placed matters as much as *how many*. Two decades later, HRSA still tracks grants at the organization level, not the site level, with no public tools to evaluate whether funding reaches the neighborhoods with the greatest need. Pérez et al. [21] found through optimization modeling that more funding does not always lead to better health outcomes: more sites do not help if they are in the wrong places.

People in poor, predominantly Black neighborhoods on Chicago’s South Side have a life expectancy 30 years shorter than those in wealthy neighborhoods nine miles away, the largest such gap in the United States [24]. Luo et al. [17] showed on the same South Side that better spatial accessibility to primary care is associated with lower hypertension prevalence. Lee et al. [16] studied six states including Illinois and found that in MUAs with at least one FQHC, patient volume was 3.27 times higher than in MUAs with none, and that FQHC placement patterns may reflect historic redlining.¹ These patterns suggest a long-standing mismatch between where facilities are placed and where they are most needed.

In summary, Gap 1 is addressed by the district and superdistrict design in Section ??, which lets decision makers evaluate candidate locations against existing capacity, demand, and a user-defined IMU-like score. Gaps 2 and 3 are addressed by the redesigned IMU-like scores in Section 2.2. Gap 4 is addressed at the superdistrict level through hospital referral relationships, and more fully in the follow-up study proposed in Section 4.

2 Model Formulation

The problems described above involve making spatial decisions under constraints such as budget, capacity, travel time, equity, with many interacting components. We propose a spatial optimization framework that addresses the HRSA requirements listed in Section 1.

2.1 A hierarchical network design via FalCom

Geographic disparities in healthcare access have been studied across multiple service domains. In organ allocation, Gentry et al. [11] used integer programming to redistrict U.S. liver allocation regions for equity, and Akshat et al. [1] extended this with heterogeneous donor circles that equalize supply-to-demand ratios under contiguity and capacity constraints. The HRSA/FQHC setting we study shares the same structural form: supply and demand on a fixed geographic graph with capacity, contiguity, and equity constraints, where boundary design is itself a policy lever.

FalCom [14] is a Markov chain Monte Carlo method designed for hierarchical and capacitated facility location and districting problems. The energy of a district measures how well a facility within that district serves its demand points. The system can be designed in a way that energies (service qualities) across districts are equalized as much as possible. The sampler favors configurations

¹Redlining was a federal housing policy from the 1930s where the Home Owners’ Loan Corporation (HOLC) graded neighborhoods A through D. Banks refused loans in those areas for decades. The same neighborhoods received less investment in infrastructure, healthcare, schools, and housing for generations. Lee et al. [16] found that FQHC placement patterns today still correlate with these historical grades, meaning formerly redlined neighborhoods remain underserved.